Smart Medicine Packaging Scanner & AI Optical Character Recognition

Automated Barcode Decoding, Dual-Engine Text Extraction, & Dynamic Risk Scoring

Target Audience: Hospital Pharmacists, Inventory Managers	**Engine:** HTML5-QRCode + Tesseract.js v5 + Regex
Application: Priority-Based Expiry Risk System	**Deployment:** Vercel Serverless WSGI / Local Flask

1. Executive Overview & Clinical Rationale

In hospital pharmacies and commercial medical stores, manual entry of incoming medicine batches is a major operational bottleneck. A typical pharmacist must accurately key in 10+ data points per box: proprietary name, active chemical salt, batch number, manufacturer, unit price, stock quantity, manufacturing date, and expiry date. Transcription errors frequently lead to misrecorded expiry dates, resulting in either premature write-offs or catastrophic dispensing of expired drugs.

The **Smart Medicine Scanner & OCR Engine** automates batch ingestion. Pharmacists point their webcam or smartphone camera at packaging to instantly decode barcodes, recognize printed label typography, parse expiration dates, and automatically populate registration forms in under 1.5 seconds.

2. Dual-Engine Architecture & Processing Pipeline

The system employs a multi-tiered pipeline that blends high-speed client-side computer vision with serverless regex parsing:

Pipeline Stage	Component	Latency	Functional Responsibility
Stage 1: In-Memory Downscaling	HTML5 Canvas 2D	< 20 ms	Shrinks 12MP photos to 950px max dimension. Applies grayscale and high-contrast edge thresholding (25× pixel reduction).
Stage 2: Barcode Decoding	HTML5-QRCode (ZXing)	15–35 ms	Extracts EAN-13, UPC-A, Code-128, and QR barcodes. Queries SQLite database immediately to check for duplicate batches.
Stage 3: Optical Character Recognition	Tesseract.js v5 (WASM)	0.8–1.5 s	Pre-warmed background WebAssembly worker extracts printed typography from packaging (Brand, Salt, Dates, Batch, Price).
Stage 4: RegEx Date Parsing	Python Date Engine	< 1 ms	Normalizes varying pharmaceutical date conventions (YYYY-MM-DD, MM/YYYY, DD/MM/YYYY, EXP: MMM-YY) into ISO standards.

Stage 5: Dynamic Risk Scoring	Priority Risk Engine	< 1 ms	Computes 0–100 risk score based on remaining shelf life, stock volume, unit price, and monthly demand velocity.

3. Operational Modes & User Interfaces

The scanner is accessible via multiple entry points tailored to diverse clinical workflows:

A. Standalone Full Scanner Page (/scanner)

- **Live Camera Viewfinder:** Real-time video stream running at 15 FPS with automatic barcode capture. Features an instant *'Snap Photo & Read Label'* button for full OCR extraction.
- **Dedicated File Upload Mode:** Allows dragging and dropping high-resolution label photos, browsing device files/galleries without camera forcing, or capturing via mobile camera.
- **Verification & Review Screen:** Extracted fields are displayed with an interactive live risk preview badge, alert banners for duplicate batches, and a 1-click confirmation button.

B. Integrated Inline Scanner (/medicine/add)

- Embedded directly atop the inventory registration form, allowing pharmacists to scan without navigating away.
- Features a toggleable camera viewfinder, drag-and-drop file input, and 1-click pharmaceutical sample buttons.
- Upon scan completion, all form inputs (Name, Batch, Barcode, Dates, Price) populate automatically with a green highlight animation.

C. Inventory Toolbar Direct Access (/medicines)

- Features an eye-catching amber **'Scan Medicine'** button in the main toolbar, allowing fast access from the active stock table.
- Displays scanned barcodes directly under batch numbers in the inventory table for barcode audits.

Performance Engineering Note (25× Speedup):
Earlier versions passed uncompressed 12MP smartphone photos (4000×3000 px = 12,000,000 pixels) directly to WebAssembly OCR, causing 20–30s browser stalls. Our optimized pipeline downscales images on an in-memory canvas to 950px (~450,000 pixels) and pre-warms the Tesseract worker on page load. OCR processing time was reduced from **30 seconds to ~1.2 seconds**.

4. Standard Operating Procedure (SOP): Batch Ingestion

Step	Action	System Response	User Verification
1	Navigate to Scanner or Add Medicine	Camera feed activates; background OCR worker pre-warms.	Verify webcam permission prompt.
2	Present label or upload packaging photo	Barcodes decode instantly (<35ms). Canvas downscales image and runs OCR.	Ensure label is well-lit and unobstructed.
3	Review extracted batch attributes	Populates Medicine Name, Batch Number, Mfg Date, Expiry Date, and Price.	Confirm parsed Expiry Date against physical packaging.
4	Inspect Live Risk Score Preview	Calculates dynamic risk (CRITICAL, HIGH, MEDIUM, LOW, SAFE).	Note any immediate expiration flags.
5	Click 'Confirm & Add to Inventory'	Saves record to SQLite database and enters FEFO priority queue.	Item is now live in inventory and reports.

5. Multi-Factor Expiry Risk Formula

Once the expiry date is parsed from the packaging, the system does not simply sort by calendar date. It evaluates four distinct operational risk dimensions to compute a transparent **0–100 Risk Score**:

Risk Dimension	Weight	Scoring Matrix & Criteria
1. Expiry Proximity (S_expiry)	**50%**	Expired (≤0d): 100 pts | 1–30d: 90 pts | 31–60d: 75 pts | 61–90d: 55 pts | 91–180d: 35 pts | >180d: 10 pts
2. Stock Quantity Exposure (S_stock)	**20%**	≥500 units: 100 pts | ≥250 units: 75 pts | ≥100 units: 50 pts | ≥50 units: 30 pts | <50 units: 15 pts
3. Financial Exposure (S_value)	**15%**	Total Batch Value = Quantity × Unit Price. ≥■20,000: 100 pts | ≥■10,000: 80 pts | ≥■5,000: 60 pts | ≥■2,000: 40 pts | <■2,000: 20 pts
4. Demand Surplus Velocity (S_surplus)	**15%**	Supply Months = Quantity / Monthly Usage. Coverage > 3× shelf life: 100 pts | High surplus: 75 pts | Moderate: 50 pts | Adequate: 25 pts

6. Technical Specifications & Serverless Deployment

• **Client-Side Libraries:** HTML5-QRCode v2.3.8 (ZXing barcode core), Tesseract.js v5 (WASM worker), Bootstrap 5.3.
• **Backend Framework:** Python 3.8+ / Flask 3.x WSGI runtime.
• **Storage Engine:** SQLite3 database with automatic table creation and schema validation.
• **Vercel Serverless Ready:** Uses ephemeral /tmp database allocation and serverless upload fallback routing.
• **Hardware Compatibility:** Compatible with USB barcode guns, smartphone rear cameras (autofocus environment lens), desktop webcams, and flatbed document scanners.

Document Approved by: Lead Software Architect & Clinical Informatics	**Status:** Production Ready (v2.4)
Repository: Priority-Based Medicine Expiry Risk System	**Deployment URL:** medical-expiry-risk-system.vercel.app